[Dec-25]

GITAM (Deemed to be University)
[CSCI6131]
MCA Degree Examination

I Semester
(Supplementary)

OPERATING SYSTEMS: CONCEPTS AND DESIGN

(Effective for the admitted batch 2022-2023-2024-25

Time: 3 Hours **Max. Marks: 60**

Instructions: All parts of the unit must be answered in one place only.

Section-A

1. **Answer all Questions:** **(10×2=20M)**

   a) Illustrate the importance of OS in different perspectives.

   b) What is Open-Source Operating System?

   c) Define process and draw process state diagram.

   d) What is the purpose of medium term Scheduler?

   e) Analyze critical section.

   f) Identify the purpose of Resource Allocation graph.

   g) Analyze swapping technique.

   h) List out the design issues of paging system.

   i) Classify disk scheduling algorithms**.**

   j) Define file and directory.

Section-B

Answer the following: **(5×8=40)**

UNIT-I

2. List essential properties of the following types of Operating system:
   a) Batch operating system
   b) Interactive operating system
   c) Time sharing operating system

OR

3. What are system calls? Explain the different categories of the system calls

UNIT-II

4. Distinguish between user-level threads and kernel-supported threads.

OR

5. Explain the following with the term related to IPC:
   a) Race condition
   b) Critical region

UNIT-III

6. What is semaphore? Discuss producer consumer problem with semaphore.

OR

7. Elaborate Resource request algorithm for Deadlock avoidance.

UNIT-IV

8. Define demand paging? Analyze the address translation mechanism with a neat diagram.

OR

9. Define the term locality of reference and elaborate on its usefulness in presenting thrashing.

UNIT-V

10. Distinguish between
    a) File-System Mounting
    b) File Sharing.

OR

11. Categorize the methods of free space management of a Disk?